

Ghibli Market: LoRA Style-Tuning with Stable Diffusion v1.5

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Abstract

This project presents a parameter-efficient approach for adapting Stable Diffusion 1.5 to generate Ghibli-inspired market scenes using Low-Rank Adaptation (LoRA). A custom style token, `<sks>`, is added to the tokenizer to provide direct prompt-based control over the learned visual style, while LoRA modules are applied to both the UNet and text encoder as required by the assignment. The model is trained on the provided 512×512 reference images, with the original Stable Diffusion parameters kept frozen to reduce computational cost and preserve the base model’s capabilities. The final pipeline saves the learned adapters in a single SafeTensors file and includes an evaluation script that loads the trained weights and generates reproducible images from prompts such as “a busy market, in `<sks>` style.” The result is a compact, controllable, and reproducible style-tuning system that captures the distinctive visual characteristics of the reference dataset without requiring full model fine-tuning.

Architecture

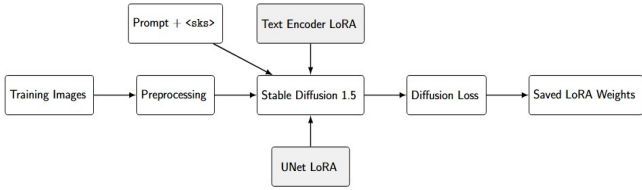


Figure 1: Overall architecture of the proposed LoRA fine-tuning pipeline.

The proposed system fine-tunes Stable Diffusion 1.5 to generate Ghibli-inspired market scenes using Low-Rank Adaptation (LoRA). Rather than updating the full model, lightweight LoRA adapters are inserted into the attention layers of the UNet and the CLIP text encoder, while the original VAE, UNet, and text encoder weights stay frozen. This substantially reduces the number of trainable parameters and the computational cost of fine-tuning. A custom style token, `<sks>`, is included in the text prompts. The LoRA adapters learn to link this token with visual characteristics of the reference dataset during training. At testing time, the token can be included in a prompt to generate new images in the learned style. The training process follows the standard latent diffusion framework. Images are encoded into latent representations, noise is added at randomly selected diffusion timesteps, and the UNet predicts the added noise under text conditioning. The LoRA adapter parameters are updated using the noise-prediction loss. Once training is done, the compact adapter weights are saved and loaded alongside the original Stable Diffusion model for image generation.

Training Pipeline

During training, the Stable Diffusion 1.5 components were loaded separately: tokenizer, CLIP text encoder,

VAE, UNet, and DDPM noise scheduler. The custom token `<sks>` was added to the tokenizer, after which the text encoder embedding matrix was resized to include the new token. To provide a meaningful starting point for the new token, its embedding was initialized using the embedding of the existing word “style”.

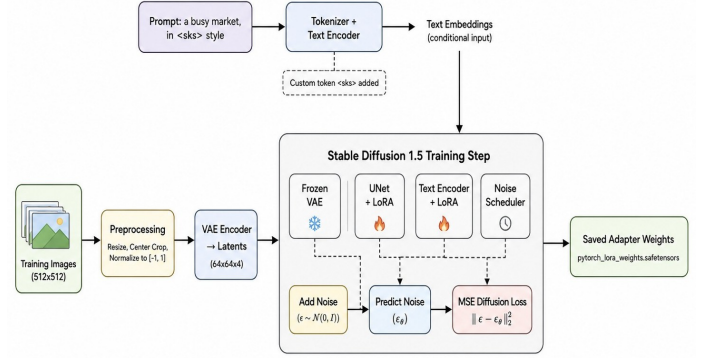


Figure 2: Training pipeline used for LoRA fine-tuning of Stable Diffusion v1.5.

The base VAE, UNet, and text encoder parameters were frozen before applying LoRA. LoRA adapters were then attached to both the UNet and text encoder, satisfying the assignment requirement for dual-adapter LoRA fine-tuning. For the UNet, LoRA was applied to the attention projection layers `to_q`, `to_k`, `to_v`, and `to_out.0`. For the text encoder, LoRA was applied to `q.proj`, `k.proj`, `v.proj`, and `out.proj`. The original text encoder parameters remained frozen throughout training.

The training objective followed the standard diffusion noise-prediction setup. Each input image was encoded into latent space using the VAE, random Gaussian noise was added at a randomly sampled diffusion timestep, and the UNet was trained to predict the added noise conditioned on the text encoder output. The loss function was mean squared error between the predicted noise and the true noise. AdamW was used as the optimizer. The final trained adapter was saved as a single LoRA SafeTensors file, containing the UNet LoRA weights, text-encoder LoRA weights, and the learned embedding of the custom `<sks>` token.

Hyperparameters

The final model was trained using Stable Diffusion v1.5 on 843 images. A LoRA rank of 16 was selected to provide greater adaptation capacity than the suggested rank of 8 while keeping the number of trainable parameters substantially lower than full fine-tuning.

A learning rate of 1×10^{-4} was used, as it provides stable optimization for LoRA fine-tuning and letting the LoRA adapters to learn the new style efficiently. Batch size was set to 1, mainly due to GPU memory constraints when training Stable Diffusion at full image resolution.

Table 1: Hyperparameters used for LoRA fine-tuning

Hyperparameter	Value Used
Base model	runwayml/stable-diffusion-v1-5
Dataset size	843 images
Style token	<sks>
Image resolution	512 × 512
LoRA rank	16
Learning rate	1 × 10 ⁻⁴
Batch size	1
Maximum training steps	1600
Checkpoint interval	200 steps
Random seed	42

Training was performed for 1600 optimization steps, increasing from the suggested 800 steps to evaluate whether longer training improved the consistency of the style. Checkpoints were saved and the random seed was fixed to 42 to improve reproducibility. The final adapter weights were saved in SafeTensors format.

Results

Fine-tuning Stable Diffusion v1.5 with LoRA proved to be an effective approach for adapting the model to generate Studio Ghibli-inspired illustrations. The model was trained on a dataset of 843 images for 1600 optimization steps with a LoRA rank of 16. Checkpoints were saved every 200 optimization steps during training and evaluated using the prompt “a busy market, in <sks> style”. This enabled a qualitative comparison of the generated images at different points.



Figure 3: Qualitative comparison of images generated from LoRA checkpoints after 1000, 1200, 1400, and 1600 training steps using the prompt “a busy market, in <sks> style.” The generated images show progressively stronger Ghibli-inspired characteristics, with the 1600-step checkpoint producing the most consistent style and highest visual quality.

Figure 3 illustrates representative outputs generated from the checkpoints at 1000, 1200, 1400, and 1600 training steps.

The earlier checkpoints generated realistic market

scenes with relatively weak stylization, and the later checkpoints show increasingly stronger Ghibli-inspired features. The last checkpoint with 1600 steps achieved the best trade-off between preserving the market content, painterly textures, and more consistent Ghibli-inspired illustrations.

To further improve image quality, different inference configurations were evaluated by varying the number of denoising steps and guidance scale. Based on the qualitative evaluation, the best results were achieved with 50 inference steps with a guidance scale of 7.5. This configuration provides images with finer details, more visually coherent characters, and a better overall balance between style and content compared to the other settings.

Figure 4 compares the base Stable Diffusion v1.5 model with the proposed LoRA model after 1600 training steps. While the base model generates a realistic market scene, the fine-tuned model produces a consistent Ghibli-inspired artistic style while preserving the market content. The learned <sks> token successfully controls the visual style, demonstrating that the LoRA adapters effectively capture the characteristics of the reference dataset. Although minor artifacts remain in some facial details and small objects, the overall image quality and stylistic consistency are substantially improved.

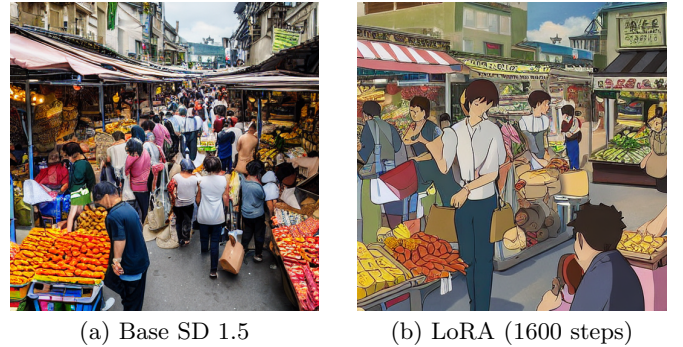


Figure 4: Comparison between the base Stable Diffusion v1.5 model and the proposed LoRA model after 1600 training steps using the same prompt. The LoRA model produces a consistent Ghibli-inspired style while preserving the market content.

Conclusion

This project demonstrated that the use of Low-Rank Adaptation (LoRA) can efficiently adapt Stable Diffusion v1.5 to a new artistic domain. By training LoRA adapters on both the UNet and CLIP text encoder while keeping the original model weights frozen, the model successfully learned a Studio Ghibli-inspired style from a relatively small dataset.

The experimental results showed that the later checkpoints showed the consistency of the learned style. Among the evaluated checkpoints, the 1600 steps checkpoint provided the best overall balance between style consistency and prompt fidelity. Furthermore, evaluating different inference settings revealed that image quality is influenced not only by the trained model but also by the sampling parameters. Overall, the results demonstrate that parameter-efficient tuning is a solid approach for adapting large diffusion models to specific artistic domains. Future work could explore larger and more diverse training datasets and different hyperparameter settings to further enhance style consistency and image realism.